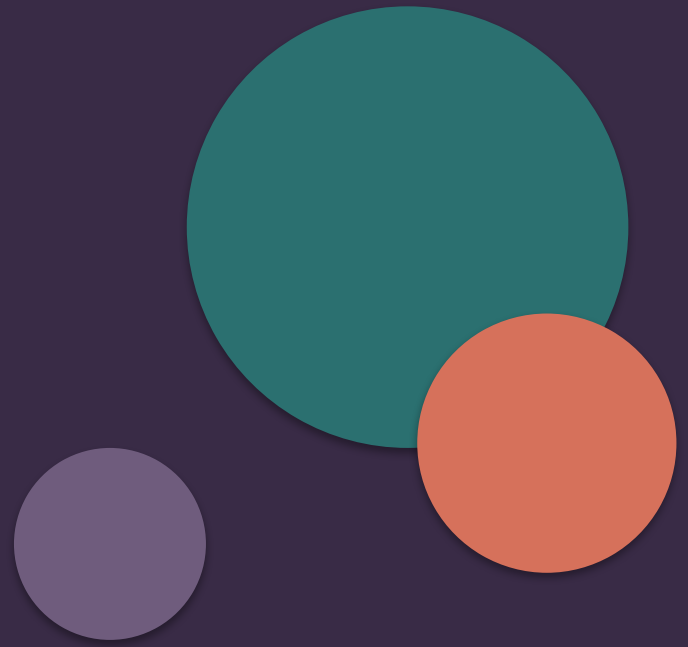
DDCA • CO4

8-bit Signed Binary Representation

Understanding signed numbers through sequential logic concepts



Latches • Flip-flops • Registers • Counters

What does ‘8-bit signed’ mean?

An 8-bit pattern contains eight binary digits; signed representation lets it express positive and negative integers.

8-BIT PATTERN

```
0101 1010
```

Key idea

The same 8 bits can represent a number according to the chosen signed encoding.

For modern examples, we usually use two’s complement.

Bit count

8 bits = 1 byte

MSB

Leftmost bit; sign indicator in two’s complement

LSB

Rightmost bit; contributes \(2^0\)

Two’s-complement range

For n bits, the range is -2^{n-1} to $+2^{n-1}-1$.

8-BIT SIGNED RANGE

−128 to +127

Why is it asymmetric?

There is one more negative value than positive values: −128 exists, but +128 does not.

Smallest value

1000 0000 = −128

Largest value

0111 1111 = +127

How to encode a negative number

To form −x in two's complement: write +x in 8 bits → invert every bit → add 1.

1. Start with +5

0000 0101

2. Invert the bits

1111 1010

3. Add 1

1111 1011 = −5

The final result is still exactly 8 bits. Discard any carry beyond the leftmost bit.

Why CO4 topics matter

Sequential circuits store and update bit patterns over time; signed representation determines how those patterns are interpreted.

Flip-flops

A flip-flop stores one bit. Eight flip-flops can store an 8-bit pattern.

8-bit register

A register groups 8 stored bits into one byte-sized value.

Clocked update

At the active clock edge, the stored pattern can change to a new value.

The circuit stores bits; the signed-number convention gives those bits their numeric meaning.

Worked examples

Interpret each 8-bit pattern as two’s complement.

Positive

0000 1100 = +12

Negative

1111 0100 = −12

Zero

0000 0000 = 0

Tip: If the leftmost bit is 0, the two’s-complement value is non-negative. If it is 1, the value is negative; invert the bits and add 1 to find its magnitude.

Quick practice

Try these before checking the answers.

- 1. What is the 8-bit signed range?
- 2. Convert +9 to 8-bit binary.
- 3. Find the two’s-complement form of −9.
- 4. How many flip-flops are needed to store 8 bits?

Answers

1. −128 to +127
2. 0000 1001
3. 1111 0111
4. Eight

Key takeaways

Remember the number format and the storage hardware separately.

- 8-bit two’s-complement range: −128 to +127.
- Positive numbers use ordinary binary with leading zeros.
- Negative numbers are formed by invert + add 1.
- Flip-flops store bits; registers group bits; clock edges control updates.
- The leftmost bit indicates the sign in two’s complement.

Thank You

Sequential Logic | 8-bit Signed Binary Representation